KapitBayan system demo

A role-by-role presentation flow with email, push, and service fallbacks | 18–22 minutes

Purpose: show how DILG oversight, barangay processing, and resident web/mobile services connect. Staff make service decisions; external email and push delivery are best-effort.

Run the demo in this order

TIME	ROLE / AREA	SHOW
1 min	**Opening**	Purpose, user roles, and system path
2 min	**DILG**	Barangay onboarding, account support, analytics
5 min	**Barangay admin**	Residents, documents, queue, communications
4 min	**Resident web**	Requests, bulletin, help, profile
4 min	**Resident mobile**	Queue, documents, schedule, notification center
2 min	**Field official**	Queue ticker, clearance validation, reports
2–4 min	**Fallbacks**	Password reset, email, push, connectivity

System at a glance

CLIENTS	API AND DATA	DELIVERY
React web for residents, barangay staff, and DILG; Flutter app for residents and field officials.	Django REST API with PostgreSQL on Neon. Cloudinary stores configured media.	Brevo SMTP sends email. Firebase Cloud Messaging can send mobile push. Both depend on current provider and device settings.

Before people arrive

- Confirm the site, API, and APK point to the intended environment and recent release.
- Prepare separate synthetic DILG, barangay-admin, resident, and field-official accounts.
- Pre-open separate browser profiles and sign the resident phone in; only enable push for a real phone demo.
- Verify a controlled test email and Brevo sender. Keep passwords, reset links, resident evidence, and provider secrets off screen.
- Have a clearly labelled screenshot or recording ready if the network or provider is unavailable.

1. DILG and barangay administration

Move from cross-barangay oversight to the work performed inside one barangay.

DILG administrator | about 2 minutes

- Barangay Applications: review an application and show approval/rejection and initial staff-account setup.
- Account Support: show scoped lookup, verified email correction/recovery, account activation status, and the audit trail. This is bounded recovery support, not password viewing or unrestricted deletion.
- Global Analytics: show cross-barangay summaries.

> Use a demo account. Do not alter a real deployed account or send a recovery email during the demo unless the account owner requested it.

Barangay administrator | about 5 minutes

Stay inside one demo barangay to make the access boundary clear.

Residents and verification

- Open Residents and show the account record and verification review. Registration does not equal approval; a pending resident cannot use protected services until staff verify the account.

Documents and treasury controls

- Review a document request, record a fee or supported waiver, and show a status change. Payment evidence is reviewed manually; show the Official Receipt controls where relevant.

Queue and resident paging

- Add or select a ticket, advance the queue, and show the serving ticket, waiting list, and public display.
- Use Page resident only for the current serving ticket. This saves an in-app alert, then separately attempts email and FCM according to the resident's preferences.

Communications and service work

- Create or edit announcements and activity schedules; maintain hotline entries and resident FAQs. Show the published resident view.
- Open an assistance request or incident report and show the staff update. Staff decide outcomes; the system does not automate aid eligibility or emergency response.

Staff notifications

- Show the staff notification center and read/clear actions. It is separate from the resident's mobile notification inbox.

> Electronic transfer references and uploaded receipts are evidence for staff review, not proof of settlement. KapitBayan does not process payments through a gateway.

2. Resident and field workflows

Show the same service from the resident side, then switch to the supported field role.

Resident web portal | about 4 minutes

- Show registration and the pending-verification state. Explain that barangay staff approval is required before protected login.
- Submit a document request and follow its status. If payment is available, point out that the staff fee assessment and barangay payment settings control the next step.
- Open Queue, Community Bulletin, and Aid. Show announcements and activities published by the barangay.
- Open Help / FAQs, Tutorial, and Profile. Residents can return to help at any time; the tutorial uses short steps, including guidance useful to older residents.
- Show the email notification preference. The web portal does not register an FCM device; mobile push is configured in the app.

Resident mobile app | about 4 minutes

- Show the dashboard, Queue, Documents, and Schedule. Then show Announcements, Aid, Issue Reports, and Hotlines as available in the current build.
- In Profile → Notification Preferences, show the push setting. Push requires the app preference, Android permission, a registered token, network access, and matching Firebase projects.
- Open the notification center to show saved queue paging alerts. If the hotline button cannot launch the phone app, show the number and dial it manually.
- Open Help / FAQs and Tutorial. Use the candidate APK; an older APK may not contain recent notification changes.

Field official mobile workflow | about 2 minutes

- Queue ticker: view queue status and permitted field actions.
- Clearance validator: enter a tracking ID and inspect the returned status.
- Field reports: review barangay incident reports and record supported updates. Field actions are limited by role and barangay.

> The mobile app requires network access for server-backed workflows. If the service is unavailable, use the barangay desk or official hotline; confirm status before sending a duplicate request.

3. Password recovery and Brevo email

Demonstrate only with a test account and an inbox you control.

STEP	PRESENTER ACTION	EXPLAIN
1	Choose Forgot password and submit a test address.	Use an address associated with exactly one account record.
2	Show the generic success message.	It protects account privacy; it does not confirm an email was sent.
3	Check Inbox and Spam.	If missing, inspect Render logs and Brevo Transactional email logs.
4	Use account support when needed.	An authorized DILG operator verifies the holder and destination before requesting a private recovery link.

Current Render email configuration: Brevo SMTP relay at smtp-relay.brevo.com, port 2525 with TLS, Brevo SMTP login/key, and a verified sender. Never show the key or an account password.

If no reset email arrives

- The endpoint dispatches only when exactly one user record matches the address. Unknown or duplicate addresses can get the same generic response without a message.
- HTTP 429 means the request is throttled. Wait for the window to expire, then retry once.
- No Brevo event: inspect API dispatch logs and the account match. SMTP error: check host, port 2525, TLS, SMTP login/key, and verified sender in Render.
- Brevo says delivered: check Spam, filters, and the recipient address. Follow the event reason if it says bounced or blocked.

Email facts to say out loud

- The optional service-update email preference does not disable password-reset email.
- The request response and a healthy API status do not prove provider delivery.
- Email and push run as best-effort background work; there is no durable retry queue.

For details, use the repository guide: docs/walkthrough/brevo-email-setup.md

4. Notifications, fallbacks, and close

Demonstrate the queue paging path, then explain how to recover when one channel is unavailable.

Barangay staff pages a serving resident → the backend saves a resident-owned queue alert → email and FCM are attempted separately when enabled.

CHANNEL	WHAT THE RESIDENT SEES	REQUIREMENTS / LIMIT
In-app queue alert	Saved paging message in the mobile notification center.	Use this as the fallback when no OS push banner appears; API access is still required.
FCM mobile push	Foreground in-app banner; Android may show a system notification in the background.	Mobile opt-in, OS permission, valid token, network, and server/APK Firebase project match.
Email	Optional service-update email to the resident's address.	Email preference, address, active provider configuration, and best-effort send.

If the push banner does not appear

- Open the app notification center and refresh; a saved queue paging alert can remain available there.
- Confirm Profile push is enabled, Android permission is allowed, the phone is online, and the current app is registered.
- UnregisteredError: Firebase invalidated an old token; sign in and enable push again to register a new one.
- SenderIdMismatchError: the operator must compare the Firebase project in the server service account with the mobile APK.
- If the in-app alert is missing too, check the resident notification API, database migration, and backend logs.

Other simple fallbacks

- Resident still pending: verify the same account and barangay approval; do not bypass review.
- Queue or form fails: check connection, refresh, and confirm status before resubmitting.
- Phone dialer does not open: copy or read the hotline number and dial from the device.
- Network/provider unavailable: use the barangay desk, public queue display, or official hotline. Label any recording as a backup, not a live result.

Close: “KapitBayan connects resident requests, barangay processing, and queue visibility. Staff retain service decisions, payments are manually verified, and email and push depend on provider and device settings. The demo does not replace official barangay channels.”

Presentation reminders

Use synthetic records. Do not promise guaranteed delivery or full offline operation. Verify the environment and app version before the live demo.